Diseases of Gerbera

1. Anthracnose - *Colletotrichum gloeosporiodes*
2. Phyllosticta leaf spot - *Phyllosticta gerbericola*
3. Powdery mildew - *Erysiphe cichoracearum* & *Oidium erysiphoides f. sp. gerberae*
4. Ascochyta leaf spot - *Ascochyta gerberae.*
5. Blossom blight/stalk rot /grey mold - *Botrytis cinerea*
6. Foot and root rot - **foot rot** -*Phytophthora cryptogea*
   root rot - *Pythium irregulare, Rhizoctonia solani*
7. Bacterial blight - *Pseudomonas cichorii*
8. Mosaic - *Cucumber Mosaic Virus, Gerbera Mosaic Virus*
9. Chlorotic mottle -

Anthracnose - *Colletotrichum gloeosporiodes*

Symptoms

- The diseases has been reported from Karnataka and Maharashtra.
- The disease appears as **circular, scattered, reddish brown spots.**
- They coalesce with one another during moist weather involving large area result in withering, rolling and drying of leaves.

Anthracnose - *Colletotrichum gloeosporiodes*

Management

- Excessive watering and crowding of plants should be avoided.
- Diseased leaves should be collected and burnt.
- Fungicidal spraying

Phyllosticta leaf-spot - *Phyllosticta gerbericola*

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Symptoms

- *P. gerbericola* spots are mostly circular to irregular, intially light buff later turn dark saffron, then to brown with dark brown margin.
- Spots are scattered on the lamina and **have smooth margins.**

Phyllosticta leaf spot - *Phyllosticta gerbericola*

Management

- Use certified seed
- Sanitation
- Disease-free seed and seedlings
- Collect and destroy the crop residue after harvest
- In case of heavy infestation spraying copper oxychloride (0.25%)
- Spraying of Bavistin (0.1%) or Dithane Z-78 (0.2%).

Powdery mildew - *Erysiphe cichoracearum & Oidium erysiphoides* f. sp. *gerberae*

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Photo B. Powdery mildew on gerbera leaf.

Photo A. Powdery mildew on gerbera flower.

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- This disease can affect all parts of the plant, commonly prevalent in polyhouses, considerable losses and considered as major production constraint.
- Reduced plant growth and lower plant quality contribute to economic losses due to this plant disease
- Powdery mildew is easy to identify since noticeable **white spots or white patches appear on the upper and lower surfaces of the leaves**
- These spots gradually enlarge to form a **white, powder-like mat that can spread to flowers and stems**
- **Severely infected leaves turn pale yellow or brown** and eventually die.

Powdery mildew - *Erysiphe cichoracearum & Oidium erysiphoides* f. sp. *gerberae*

Pathogen

- The fungus is an obligate parasite; it can only infect living tissue.
- The spots are a combination of body of the fungus thread like (hyphae) and upright structures, the conidiophores, which bear the conidia
- Powdery mildew reproduces prolifically by spores (conidia) , which can germinate in the absence of free water and are spread by air currents and water splash.

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Management Practices: Cultural Control

- Only plantlets that are free of powdery mildew should be used, or the plantlets should be disinfested before transplanting.
- In addition, plants should be grown in sunny areas with good air ventilation and without excessive fertilization.
- Overhead irrigation may reduce powdery mildew since it washes conidia from the leaf surfaces.
- However, **overhead irrigation may contribute to other disease problems, such as crown rot**

Powdery mildew - *Erysiphe cichoracearum & Oidium erysiphoides* f. sp. *gerberae*

Chemical Control:

- Foliar application of Triadimefon, Kresoxim-methyl, Azoxystrobin, to protect plants
- Trifloxystrobin, myclobutanil, Azoxystrobin, Potassium bicarbonate, and chlorothalonil + thiophanate-methyl are effective for the management of Powdery mildew.
- However, chemical control may not always be completely effective since the fungus may develop resistance to some chemicals.
- Since powdery mildew goes through many short cycles per season, the potential for development of resistance is great, especially when only systemic fungicides are used to control this disease.
- **Resistance of powdery mildew has been confirmed with the fungicides and triadimefon (Bayleton) in the past.**

Ascochyta leaf spot - *Ascochyta gerberae*

Blossom blight/stalk rot/grey mold - *Botrytis cinerea*

Symptoms:

- Dead areas occur on **leaves, flowers and stems of infected plants**
- The diseased leaf areas enlarge rapidly under favourable conditions and eventually entire leaves **may turn black and die.**
- The **infected leaves and flowers** become covered with a brownish gray mold consisting of masses of fungal spores
- All **old leaves, flowers and diseased parts** and plants should be removed.
- **Wider spacing to improve air circulation**

Blosoom blight/stalk rot/grey mold - *Botrytis cinerea*

Management:

- Space plants to ensure good air circulation so as to minimise disease incidence
- Maintain low humidity.
- Avoid watering late in the day. Remove crop debris.
- Apply 0.2 % Chlorothalonil or 0.1 % Azoxystrobin or Fludioxonil or Thiophenate methyl to protect healthy plants

Crown and Root Rot - *Phytophthora cryptogea*

Symtoms

- Plants fail to grow adequately and **remain stunted.**
- The **foliage develops** an obvious off-color.
- **Petioles may become infected** where they attach in the **crown area, as the petiole collapses**, the attached leaf also die.
- As the disease progresses, **the entire plant wilt and die.**
- *Phytophthora* root and crown results in a rapid plant wilt, in which the plants appear normal and healthy one day but develop symptoms of rapid wilt and decline.

Crown and Root Rot - *Phytophthora cryptogea*

Symtoms contd….

- The pattern in the flat or on the greenhouse bench is usually random but can be extensive, involving a large number of plants, especially if plants are not grown on benches.
- Overhead watering readily splashes the pathogen from plant-to-plant.
- Examination of the symptomatic plant will usually reveal a very wet root ball with significant root and crown discoloration and deterioration

Foot and root rot - *Pythium irregulare, Rhizoctonia solani*

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- It is a soil borne disease caused by *Pythium irregulare* and *Rhizoctonia solani* which are soil borne pathogens.
- Plants wilt and die as roots rot

Foot and root rot - *Pythium irregulare, Rhizoctonia solani*

Management

- Soil sterilization gives good control of the disease.
- Plant in sterilized potting media.
- Apply etridiazole or etridiazole + thiophanate methyl to protect healthy plants
- This disease can be controlled by soil sterilization with vapam at 100 ml/m^2.

Bacterial Leaf blight - *Pseudomonas cichorii*

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- It was recorded for the first time in Victoria in 2008 on Gerbera.

- *P. cichorii* can infect a wide range of hosts, basil, cabbage, celery, chrysanthemum, cyclamen, endive, *Hibiscus impatiens, Vinca rosea,* lettuce and pepper

Symptoms

- Small to large spots are circular at first, then become irregular and dark brown to black.

- May have a concentric ring pattern.

Bacterial Leaf blight - *Pseudomonas cichorii*

Management

- Use disease-free seed.
- Crop rotation with non-host crops for at least 2 years
- Destruction of plant debris
- Deep ploughing
- Balanced nitrogen fertilisers application.
- Application of bactericides

Gerbera Mosaic - *Cucumber Mosaic Virus, Gerbera Mosaic Virus*

Symptoms:

- Leaves may be mottled, have yellow spotting, dead flecks, line or ring spot patterns.
- Plants may be distorted or have flower colour breaking.
- *Cucumber Mosaic Virus, Gerbera Mosaic Virus*
- The virus was transmitted non persistently by *Myzus persicae* and *Aphis gossypii*